$$\boldsymbol{v}^T M' \boldsymbol{v} \leq \alpha^2 - 4\alpha^2 + \underbrace{|A|(|A|-1)\frac{\alpha}{|A|^2}}_{\leq \alpha} + \underbrace{\frac{1}{|A|}}_{\leq \alpha^2} + \underbrace{|B|(|B|-1)\frac{\alpha}{|B|^2}}_{\leq \alpha} + \underbrace{\frac{1}{|B|}}_{\leq \alpha^2} - 2\alpha \leq -\alpha^2 < 0,$$